

Class: XI 'Science'

MORNING SHIFT 'SET-B'

F.M: 37

Time: 1.5 hrs.

Subject: Physics [1011]

P.M: 15

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempts all the questions.

GROUP - 'A' [6X1=6]

Multiple Choice Questions:

1. The equation of a wave is given by, $y = A \sin \left[\omega \left(\frac{x}{v} - k \right) \right]$, when ω is angular velocity x is position and v is linear velocity. The dimensional formula for k is:
a) $[LT]$ ☒ b) $[T]$ c) $[T^{-1}]$ d) $[T^2]$
2. Identify the correct statement:
☒ a) Velocity, displacement, acceleration, displacement are vectors
b) Weight, mass, torque and energy are scalars
c) Angular velocity, momentum and electric current are scalars
d) A vector quantity is one which has magnitude only whereas scalar quantity is one which has both magnitude and direction.
3. When 2×10^{13} electrons are added to a neutral metal sphere, the charge on the sphere becomes:
a) $-16 \mu C$ b) $16 \mu C$ c) $32 \mu C$ d) $-32 \mu C$
4. A soap bubble is given a negative charge, then its radius:
a) Decreases b) Increases c) Remains unchanged
d) Nothing can be predicted as information is insufficient.
5. The force between two charges 0.06 m apart is 5 N. If each charge is moved towards the other by 0.01 m, then the force between them will be
a) 7.20 N b) 11.25 N c) 22.50 N d) 45.0 N
6. When a positively charged, conductor is connected to earth:
a) No charge flow occurs
b) Electrons flow from conductor to earth
c) Protons flow from conductor to earth
d) Electrons flow from earth to conductor

GROUP - 'B' [5X3=15]

Short Answer Questions:

7. a) What are the basic rules of dimensional analysis?

[1]

- b) Can we tell the unit of a physical quantity from its dimensions? What about dimensions from the unit of a physical quantity? **NO** [1]
- c) Check the correctness of the physical equations: (i) $t = 2\pi \sqrt{\frac{l}{g}}$; t is time period, l is length and g is acceleration due to gravity (ii) $T^2 = \frac{4\pi^2 a^3}{GM}$; T is time period, G is gravitational constant, M is mass and a is radius of the orbit. [3]
8. a) What is meant by resultant vector. **Defn** [1]
- b) Two vectors acting on a point and are represented by two sides of a triangle taken in order. How can you find the resultant force using this triangular representation? State and explain the law involved. [3]
- c) Under what condition is the magnitude of resultant vector is zero vector? **equal and opposite** [1]
9. a) Explain about Action of point with a demonstration? [2]
- b) What do you mean by electrostatic induction? How will you give permanent positive charge to a conductor by the method of induction? Explain in detail. [3]

GROUP - 'C' [2X8=16]

Long Answer Questions

10. a) What do you mean by dimension of a physical quantity? What is the difference between dimensions and unit of a physical quantity? [2]
- b) A body moving through air at a speed v experiences a retarding force F given by the equation $F = kApv^x$, where A is the surface area of the body, ρ is the density of air and k is dimensionless constant. Deduce the value of x . [2]
- c) Convert 60 J/s into a new system of unit which has 10gm, 10 cm and 10 second as fundamental units of mass, length and time. [2]
- d) Is it possible to derive all the physical equations by the method of dimensional method? Explain. **NO** [2]
11. a) Write the formula for Coulomb's force in vector form? What is the significance while expressing in vector form? [2]
- b) What is electrical permittivity? Write its value and unit for vacuum in SI and cgs system of unit. [2]
- c) If a body gives out 10^9 electrons per second, how much time is required to get a total charge of 1 C from it? What conclusion would you draw from the result obtained? [2]
- d) Two positive point charges are at a separation of 0.3 m repel each other with a force of 1.6 N. If the sum of two charges be 10 microcoulomb, what are their separate values? [2]